

Lighthouse: A Self-Reconfiguring Sociotechnical Infrastructure for the Unforeseen Long-Tail of Urban Crisis

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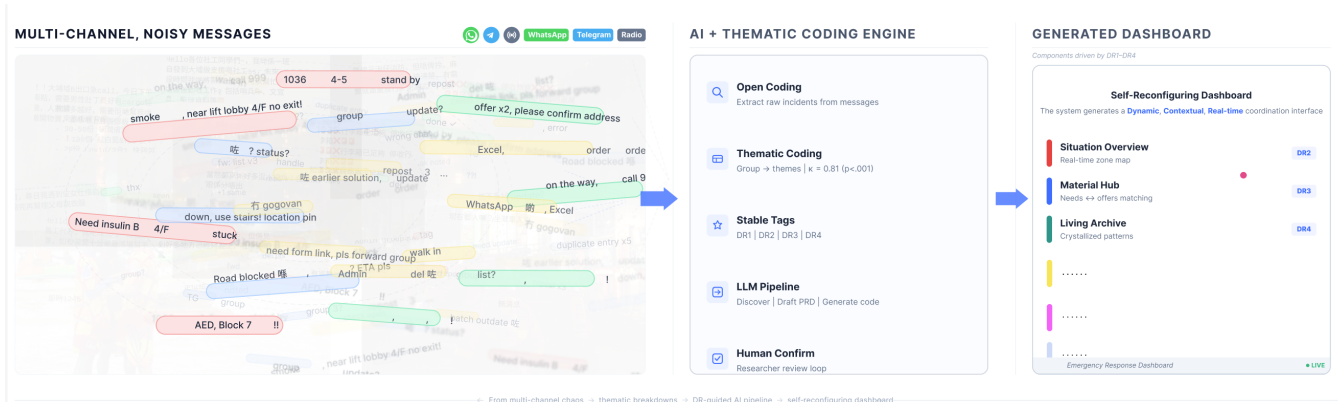


Figure 1: Lighthouse System Overview. AI-guided pipeline that transforms multi-channel, noisy crisis messages into stable thematic codes and generates a self-reconfiguring coordination dashboard for real-time situation awareness, resource matching, and pattern archiving.

Abstract

Existing crisis platforms fail because rigid schemas cannot keep pace with evolving needs (ontological drift). While Generative UI offers adaptivity, a critical research gap exists: current paradigms prioritize *individual personalization*, leaving its application to **collective resource allocation** unexplored. We introduce **Lighthouse**, a generative middleware that bridges this gap by parsing fragmented chat streams to dynamically instantiate shared coordination dashboards. To ensure reliability, a mixed-initiative stability engine anchors generation in human verification. Uniquely, Lighthouse transcends immediate response by functioning as a **living archive**: it

crystallizes ephemeral unknown unknowns” into structured knowledge, offering communities a concrete foundation from which to evolve future disaster taxonomies based on empirical ground truths rather than static assumptions.

CCS Concepts

- **Human-centered computing** → **Interactive systems and tools; Collaborative and social computing systems and tools;**
- **Computing methodologies** → *Information extraction.*

Keywords

crisis informatics, generative UI, human-AI teaming, large language models, resilience

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1 Introduction

HCI research increasingly relies on dashboards for decision support, yet most tools are predicated on the assumption of stable information architectures [9]. In the chaotic unfolding of disasters, however, this assumption fractures. Coordination occurs in noisy, high-velocity environments where needs, roles, and terminologies evolve faster than any predesigned form can anticipate [18]. Crisis informatics scholarship frames this as a fundamental clash between *communicative fluidity*—citizens’ ability to improvise and adapt language on the fly—and *infrastructural brittleness*—technical systems’ inability to update their own structure at runtime [18]: while citizens improvise complex support networks across fragmented channels [20], technical systems remain bound by design-time taxonomies (e.g., fixed lists like “shelter” or “food”) that fail to capture the “long-tail” of emerging needs [1, 22]. Lighthouse is designed precisely to resolve this clash: its generative schema induction operationalizes the former, while its mixed-initiative stability engine guards against the dangers of the latter.

Our analysis of the 2025 Tai Po fire coordination efforts reveals how this sociotechnical gap manifests in practice. While thousands of residents utilized WhatsApp’s flexibility, critical signals (e.g., “insulin for Block X”) remained buried in unstructured chatter because no preset database category existed to capture them. Even though civic technologists deployed ad-hoc relief sites using rapid AI-assisted coding tools, these platforms still lagged 12–24 hours behind shifting ground truths.

We identify this latency as a structural failure of **Ontological Drift**—the progressive mismatch between the categories a crisis *demands* and the categories a system *can represent*—a gap that widens as novel situations outpace pre-designed taxonomies: (e.g., a shift from supply tracking to complex volunteer rostering) required a fundamental re-engineering of the interface’s logic, forcing coordinators to abandon their tools and resort to manual “repair work” via screenshots [6]. These observations regarding the brittleness of static schemas directly inform the design of **Lighthouse**, a generative middleware that shifts LLMs from passive classifiers [7] to *ontological bridges*—active agents that construct new representational categories on-the-fly rather than merely populating pre-existing ones.

Current AI approaches often fall short of addressing this structural disconnect. While Large Language Models (LLMs) excel at summarizing crisis texts [13, 23], prior work typically treats them as passive analysts that populate existing database fields [7], rather than as active agents capable of inferring new organizational structures [17]. Similarly, recent breakthroughs in Generative UI, such as LLM-driven reactive component generation [4] and conversational interface synthesis [21], focus primarily on synthesizing malleable interfaces for *individual personalization*, and adaptive interfaces in Industry 5.0 emphasize tailoring to individual operators [10], prioritizing user preference over the *collective sensemaking* required in high-stakes coordination.

Furthermore, a critical dissonance exists between archival theory and practice: while scholars envision “living archives” as open-ended, participatory infrastructures [8, 15, 16], operational dashboards remain ephemeral snapshots. They often discard the high-velocity context needed to build the structured knowledge essential for both immediate recovery and future resilience.

To address this bottleneck, we introduce **Lighthouse**, a generative middleware that operationalizes this shift. Lighthouse (1) unifies multi-channel messages into a coherent stream, (2) actively induces emergent schemas for roles and needs (e.g., discovering “night-shift drivers” as a necessary category), and (3) dynamically instantiates UI components without developer intervention.

Crucially, we argue that adaptivity must not compromise operational stability. An AI-hallucinated interface could fracture the shared reality essential for coordination [18]. Lighthouse addresses this via *stabilized generation*: combining hysteresis thresholds to filter noise with human-in-the-loop (HITL) gates. High-impact structural changes are staged as drafts for verification, ensuring the generated dashboard remains a reliable *boundary object* for diverse stakeholders. By targeting this missing layer, Lighthouse moves HCI systems from static dashboards toward *self-reconfiguring infrastructures* [19] that not only evolve in pace with urban emergencies but also crystallize ephemeral signals into a lasting, structured *living archive*—a persistent, queryable record that converts crisis-specific improvisations into reusable institutional knowledge for future events.

2 Contextual Analysis: Quantifying the Lag

To ground our design in observed coordination breakdowns, we conducted a retrospective analysis of the communication logs mentioned in the introduction. This analysis surfaces concrete patterns of *Ontological Drift* that directly inform each design decision in Lighthouse, with the primary intent of grounding and motivating design decisions rather than claiming generalizable empirical findings.

2.1 Methodology

Our dataset comprises $N = 5,200+$ messages from five volunteer coordination groups (Telegram/WhatsApp) across the first 72 hours of the 2025 Tai Po fire, encompassing both English and Cantonese messages. We utilized custom Python scripts to preprocess the raw corpus, standardizing timestamps across devices and anonymizing personally identifiable information (PII) consistent with ethical guidelines.

We then conducted a systematic thematic coding of “breakdown events”—instances where volunteers explicitly abandoned structured tools in favor of free-text improvisation—following the procedure established by Braun and Clarke [3]. Coding proceeded in two stages: (1) open coding to identify raw incidents, and (2) axial coding to group incidents into higher-order themes. To ensure rigor, two researchers independently coded a stratified random sample of 20% of all flagged events ($n = 284$ messages). Inter-rater reliability was assessed using Cohen’s Kappa ($\kappa = 0.81$, $p < .001$), indicating strong agreement [5]. Discrepancies were resolved through discussion until consensus was reached.

This process surfaced four recurring breakdown categories, each mapped to a specific design requirement (DR) for Lighthouse:

- (1) **Schema Overflow** (48% of events): Needs fell outside preset categories (e.g., “Insulin Type B” vs. generic “Medical”). ⇒ **DR1: Runtime Schema Induction.**
- (2) **Temporal Drift** (27%): Needs evolved faster than update cycles, creating stale data. ⇒ **DR2: Hysteresis-Gated Stability.**
- (3) **Silo Fragmentation** (15%): Critical supply offers and requests resided in separate groups, preventing matching. ⇒ **DR3: Cross-Channel Unification.**
- (4) **Ephemeral Knowledge** (10%): Improvised solutions vanished with the chat history, preventing reuse. ⇒ **DR4: Living Archive Crystallization.**

2.2 The “Dark Window” of Coordination

Our quantitative analysis reveals a critical temporal gap. As illustrated in Figure 2, a 12-hour lag separated the emergence of complex needs from the deployment of capable tools. Specifically, while chat logs at T+2h flooded with granular requests like “Heatech Size 20” or “Insulin Type B” ([Unmapped_Need]), official schemas only offered generic “Medical” buckets. This forced volunteers to act as “human middleware,” manually verifying data and posting screenshots, which introduced 4–12 hour delays ([Manual_Sync]). By the time a custom web app was finally re-engineered and launched at T+14h, the crisis phase had already shifted, rendering the tool partially obsolete.

This data confirms that the bottleneck is not digitization per se, but **schema rigidity**. When a novel need like “insulin” appeared, it defied existing schemas (e.g., “Food/Water”), creating a vicious cycle where structured tools became less usable as needs became more specific. Lighthouse addresses this by inductively generating the schema at runtime: the system infers emergent categories from live communication patterns, stages them for coordinator approval, and upon confirmation, dynamically renders the corresponding interface components—compressing what previously required hours of redevelopment into a single human-verified step.

3 System Architecture: The Lighthouse Middleware

To bridge the “Dark Window” identified in our analysis, Lighthouse is designed as a generative middleware. Unlike passive classifiers that populate fixed fields, it functions as an **inductive engine** that ingests raw streams from existing channels, proactively induces new schemas, and dynamically instantiates stabilized coordination tools at runtime.

3.1 Pipeline: From Noise to Structure

The system operates as a four-stage pipeline (Figure 1): First, **unified Ingestion (Middleware Layer)**: Addressing the fragmentation of volunteer groups, we prioritize **meeting users where they are**. The system normalizes heterogeneous streams (WhatsApp, Telegram, SMS) into a unified JSON format, preserving critical metadata (timestamp, source group) for provenance tracking [14]. Second, **the “AI Brain” (Micro-Agent Layer)**. We decompose sensemaking into specialized agents using GPT-4o via API. **Role**

Agent infers sender intent (e.g., Volunteer vs. Victim) using few-shot prompting. **Need Agent** extracts structured entities (Item, Qty, Urgency); see Table 2 in Appendix B. **Matching Agent** performs temporal linkage, scanning past offers to generate “Draft Match Proposals.” Upon confirmation, it triggers a **Notification Service** that pushes formatted alerts back to the original chat channels, effectively closing the sociotechnical loop. Third, **dynamic Schema Induction**. To capture the “long tail” of needs, the system acts as a living ontology builder. A new category (e.g., “Insulin”) is proposed only when a signal achieves *social consensus*—operationalized as $N \geq 3$ distinct senders independently raising the same need within a 30-minute window—ensuring that schema evolution reflects genuine collective demand rather than any single urgent voice. Below this threshold, signals are logged silently and held for monitoring. These proposals then enter a **Draft State**, preventing transient noise from altering the public schema. Lastly, **runtime UI Instantiation (JSON-Driven)**. Approved schema elements are serialized into a lightweight UI JSON specification defining type, priority, and visibility. This enables the frontend to dynamically render components (such as lists, counters, or maps) without manual code deployment, reducing adaptation time from hours to seconds.

3.2 The Stability Engine & Interface

To prevent “generative disorientation,” we implement a stability mechanism grounded in mixed-initiative principles [11].

Stability Mechanisms. We employ a three-tier defense against hallucination: (1) **Hysteresis Thresholds** ensure components only appear after sustained activity and do not vanish abruptly; (2) **Staged Verification** forces high-impact structural changes (e.g., merging categories) through a **HITL Gate** for coordinator approval; and (3) **Spatial Permanence**, maintained via a fixed three-column layout (Stream | Drafts | Overview), prevents the interface from shifting unpredictably [21].

Evolution & Crystallization. The dashboard behaves as a living organism. Active categories (e.g., “Insulin”) automatically gain screen real estate. Crucially, at the end of an event, the system crystallizes the evolved schema into a persistent JSON protocol. This turns the ephemeral “unknowns” of the current crisis into structured knowledge assets for future deployment.

Theoretical Lens. Ultimately, we design these generated components as **Boundary Objects**—shared artifacts that carry different meaning for different stakeholders yet remain recognizable enough to enable coordination across silos [18]. They serve not just as UI elements, but as stabilized artifacts that align the mental models of diverse actors (medics, drivers, officials) across disconnected silos. Concretely, this motivates our fixed three-column layout (Stream | Drafts | Overview), which preserves a stable shared reference frame even as content evolves.

4 Usage Scenario: The Coordinator’s View

The following scenario is grounded in our implemented prototype pipeline (signal ingestion, Need Agent extraction, schema induction with hysteresis, and HITL approval). Elements such as the Notification Service dispatch and the Living Archive crystallization

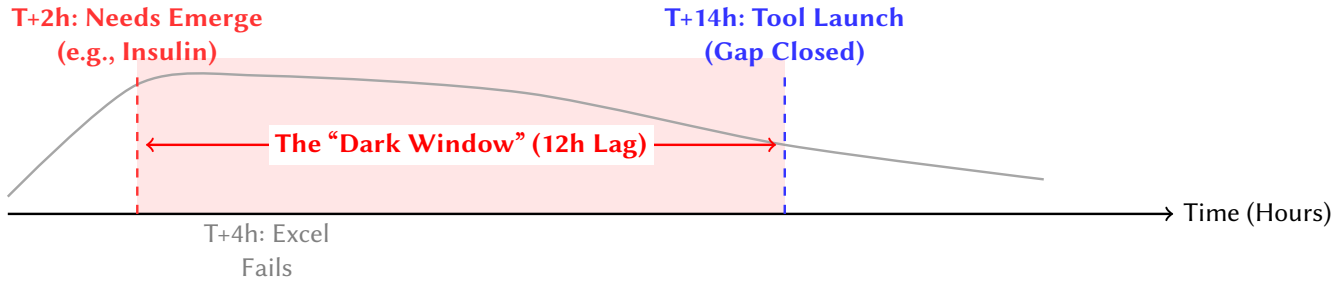


Figure 2: Infrastructure Lag Analysis. A 12-hour gap separated the emergence of complex needs (T+2h) from the deployment of capable structured tools (T+14h).

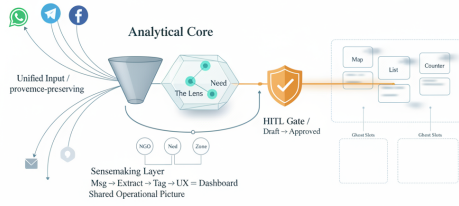


Figure 3: The Runtime Generative Dashboard visualizes these signals as a shared operational picture, featuring dynamically instantiated maps and counters.



Figure 4: The AI-driven pipeline transforms fragmented, multilingual chat streams into structured boundary objects. Real-time generative dashboard demo that unifies noisy crisis chats into a single message stream, uses an interpretable AI pipeline to infer roles, tags, and scenarios on the fly, and dynamically spawns situation-aware UI components for live coordination and resource matching.

represent the complete design vision and are subjects of ongoing validation. To illustrate Lighthouse in action, we trace a walkthrough grounded in the Tai Po Fire logs—demonstrating the core implemented pipeline from signal detection through coordinator-verified schema induction to real-time UI instantiation. We follow “Mike,” a volunteer coordinator overwhelmed by thousands of WhatsApp messages. **The Invisible Signal (Detection):** At T+2h, Mike’s standard dashboard shows generic categories like “Food” and “Water.” However, the community chat is flooding with fragmented

requests for “insulin,” “diabetic meds,” and “cold storage.” Lighthouse’s **Inductive Engine** runs in the background. Despite the code-switching and noise, it clusters these signals and identifies a pattern that defies the current schema. **The Trust Gate (Drafting):** Instead of silently changing the database, Lighthouse alerts Mike with a Draft Schema Proposal: “*New Category Detected: Cold-Chain Medication. Confidence: 0.82. Approve?*” This appears in the “Drafts” column (the HITL Gate), respecting Mike’s cognitive stability. Mike realizes this is a critical emerging need and clicks “Approve.” **Run-time Instantiation:** Upon approval, the system does not merely add a tag. It dynamically instantiates a specialized UI component on Mike’s dashboard. Unlike the generic list for water, this new widget includes a synthesized “Refrigeration Status” toggle—a field that did not exist in the database five minutes ago. **Closing the Loop (End-to-End):** With the new schema active, Lighthouse automatically scans the past 6 hours of logs. It surfaces a forgotten message from T-4h: “*I have a portable car fridge in my van.*” The system generates a Match Proposal for Mike. Once Mike confirms the match, Lighthouse acts as a bridge back to the community: it automatically dispatches a formatted notification to the original WhatsApp group, tagging both the requester and the donor to finalize the logistics, effectively completing the sociotechnical loop. **The Living Archive (Crystallization):** After the crisis subsides, the generated schema (including the new “Cold-Chain” logic) is not discarded. Lighthouse crystallizes this session into a Structured Protocol (JSON). This transforms the ephemeral “unknown unknown” of the fire into a permanent knowledge asset, ensuring that the next time disaster strikes, the system starts with “Insulin” as a known category.

5 Discussion

Design Takeaway: Lighthouse shifts the role of AI in crisis from a passive *classifier* to an active *structure generator*. This necessitates designing for **Ontological Drift**—ensuring systems can archive and adapt to the “unknown unknowns” of urban emergencies while using **Stability Engines** to prevent disorientation.

Designing for Trust in Generative Interfaces. A core tension in Generative UI is the risk of disorientation [21]. If an interface changes autonomously, users lose their spatial memory. Lighthouse addresses this by treating AI as a *proposer*, not a decider. The “Draft State” serves as a trust anchor, allowing the AI to suggest radical

structural changes (like a new map layer) without disrupting the coordinator’s workflow until explicit approval is granted. This mirrors findings from cyber situational awareness, where over-automation without accountable handoffs erodes operator trust [12]. Beyond visualization, this enables **AI-Assisted Dispatch**: by retaining a memory of past offers, the system proactively generates match proposals, reducing the cognitive load of temporal alignment.

Sociotechnical Ethics & The Power of Naming. Defining categories is an act of power [18]. If an AI fails to induce a “Wheelchair” category, that need may become invisible. Lighthouse rejects “black-box” automation through explicit governance. The “Draft & Approve” workflow forces coordinators to take responsibility for the ontology, ensuring that the schema remains a human-verified artifact and preventing the diffusion of responsibility.

Generative Archiving & Transferability. Beyond immediate response, Lighthouse functions as a “Black Box” recorder for the *epistemology of crisis*. By crystallizing ephemeral chat streams into structured schemas, the system transforms momentary noise into lasting knowledge, providing communities with a concrete, empirically grounded foundation from which to evolve future taxonomies. Furthermore, Lighthouse circumvents the “critical mass” barrier by operating as **middleware**. By ingesting data from existing high-traffic channels (e.g., WhatsApp) rather than requiring migration, it leverages existing community behaviors to populate its archives immediately, making the solution transferable to diverse urban contexts.

6 Limitations & Future Work

Evaluation Context & Bias. Our findings rely on a specific vignette (Tai Po Fire) and initial prototyping. This reflects a digitally connected context and may not generalize to low-resource settings or low-trust environments. Longitudinal deployment is needed to validate robustness.

Technical Brittleness. Current induction relies on heuristic thresholds (see Table 1). Future work should explore **few-shot learning from coordinator corrections** (e.g., learning from rejections in Table 1) to adapt thresholds dynamically, moving from static rules to personalized governance.

Future Vision: From Archive to Simulator. While this paper focuses on real-time response, the structured archive holds untapped potential. We envision using these schemas to generate realistic training scenarios and disruptive weather or cascading-failure simulations [2]. By reversing the pipeline, Lighthouse could evolve into a *resilience engine* that prepares communities for the next “unknown unknown.”

References

- [1] Abeer AlAbdulaali, Amna Asif, Shaheen Khatoon, and Majed Alshamari. 2022. Designing multimodal interactive dashboard of disaster management systems. *Sensors* 22, 11 (2022), 4292.
- [2] Anonymous. 2025. A Disruptive Weather Impact Understanding Benchmark. In *Findings of the Association for Computational Linguistics: ACL 2025*.
- [3] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology* 3, 2 (2006), 77–101.
- [4] Yining Cao, Peiling Jiang, and Haijun Xia. 2025. Generative and Malleable User Interfaces with Generative and Evolving Task-Driven Data Model. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–20.
- [5] Jacob Cohen. 1960. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement* 20, 1 (1960), 37–46.
- [6] Sara Colombo, Estefania Ciliotta, Lucia Marengo, Houjiang Liu, Piero Molino, and Paolo Ciuccarelli. 2023. Design for Emergency: How Digital Technologies Enabled an Open Design Platform to Respond to COVID-19. *Interacting with Computers* 35, 5 (2023), 650–665.
- [7] Achref Doula, Bela Bohlender, Max Mühlhäuser, and Alejandro Sanchez Guinea. 2025. CLEAR-Command: Coordinated Listening, Extraction, and Allocation for Emergency Response with Large Language Models. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (System Demonstrations)*. 20–28.
- [8] Neil Godfrey, Jayshree Mamtara, and Catherine Bow. 2016. Preserving a living archive of Indigenous language material. In *18th Biennial Conference and Exhibition. VALA*, 1–16.
- [9] Mariam Guizani. 2022. A Decade of Information Architecture in HCI: A Systematic Literature Review. arXiv:2202.13412 [cs.HC] <https://arxiv.org/abs/2202.13412>
- [10] Rania Hamdani and Ines Chihi. 2025. Adaptive human-computer interaction for industry 5.0: A novel concept, with comprehensive review and empirical validation. *Computers in Industry* 168 (2025), 104268.
- [11] Eric Horvitz. 1999. Principles of mixed-initiative user interfaces. In *Proceedings of the SIGCHI conference on Human Factors in Computing Systems*. 159–166.
- [12] Marc-André Kaufhold, Thea Riebe, Markus Bayer, and Christian Reuter. 2024. ‘We do not have the capacity to monitor all media’: a design case study on cyber situational awareness in computer emergency response teams. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–16.
- [13] Zhenyu Lei, Yushun Dong, Weiyu Li, Rong Ding, Qi R. Wang, and Jundong Li. 2025. Harnessing Large Language Models for Disaster Management: A Survey. In *Findings of the Association for Computational Linguistics: ACL 2025*. 14528–14551.
- [14] Christian Reuter and Marc-André Kaufhold. 2018. Fifteen years of social media in emergencies: a retrospective review and future directions for crisis informatics. *Journal of contingencies and crisis management* 26, 1 (2018), 41–57.
- [15] Sylvie Rollason-Cass and Scott Reed. 2015. Living movements, living archives: Selecting and archiving web content during times of social unrest. *New Review of Information Networking* 20, 1-2 (2015), 241–247.
- [16] Sarita Schoenebeck and Paul Conway. 2020. Data and power: Archival appraisal theory as a framework for data preservation. *Proceedings of the ACM on Human-Computer Interaction* 4, CSCW2 (2020), 1–18.
- [17] Philipp Seeberger, Steffen Freisinger, Tobias Bocklet, and Korbinian Riedhammer. 2025. Generalizing to Unseen Disaster Events: A Causal View. In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*. 28–37.
- [18] Robert Soden and Leysia Palen. 2018. Informing crisis: Expanding critical perspectives in crisis informatics. *Proceedings of the ACM on human-computer interaction* 2, CSCW (2018), 1–22.
- [19] Susan Leigh Star and Karen Ruhleder. 1996. Steps Toward an Ecology of Infrastructure: Design and Access for Large Information Spaces. *Information Systems Research* 7, 1 (1996), 111–134.
- [20] Kate Starbird and Leysia Palen. 2011. “Voluntweeters” self-organizing by digital volunteers in times of crisis. In *Proceedings of the SIGCHI conference on human factors in computing systems*. 1071–1080.
- [21] Ryo Suzuki et al. 2025. Generative and Malleable User Interfaces with Large Language Models. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*.
- [22] S. Xiao et al. 2025. A systematic review of social media-enabled flood disaster management: Progress, challenges, and future directions. *International Journal of Disaster Risk Reduction* (2025).
- [23] Fengyi Xu, Jun Ma, Nan Li, and Jack CP Cheng. 2025. Large language model applications in disaster management: an interdisciplinary review. *International Journal of Disaster Risk Reduction* (2025), 105642.

A System Configuration Details

Table 1 details the core hyperparameters governing the schema induction process. These thresholds were empirically tuned based on the Tai Po fire dataset ($N = 5,200$ messages).

Parameter	Value	Unit	Rationale
Cluster Activation	3	senders	Eliminates single-user noise
Temporal Window	30	mins	Captures bursts of urgent signals
Semantic Similarity	0.72	cosine	Tuned to capture paraphrases
Hysteresis (On)	0.6	score	Instantiates when confidence > 0.6
Hysteresis (Off)	0.3	score	Persists until confidence < 0.3
HITL Gate Trigger	≥ 5	fields	Significant changes require approval
Match Lookback	6	hours	Scans historical offers for matches

Table 1: Core hyperparameters for Lighthouse schema induction.

B Entity Extraction Examples

Table 2 demonstrates the Need Agent’s performance on noisy, code-switching text.

Raw Message Input	Item	Qty	Urgency	Status
“Block 7 emergency! need insulin 2 units fridge”	Insulin	2	Critical	Unfulfilled
“i have spare bottled water 10 boxes in my van”	Water	10	Low	Offered
“urgent–need driver 4am shift to central”	Driver	1	Urgent	Unfulfilled
“thanks! got the meds u sent 2 hrs ago”	Meds	–	–	Fulfilled
“anyone know where to get diapers in Tai Po?”	Diapers	–	Medium	Inquiry

Table 2: Sample extraction results. (–) denotes implicit fields.